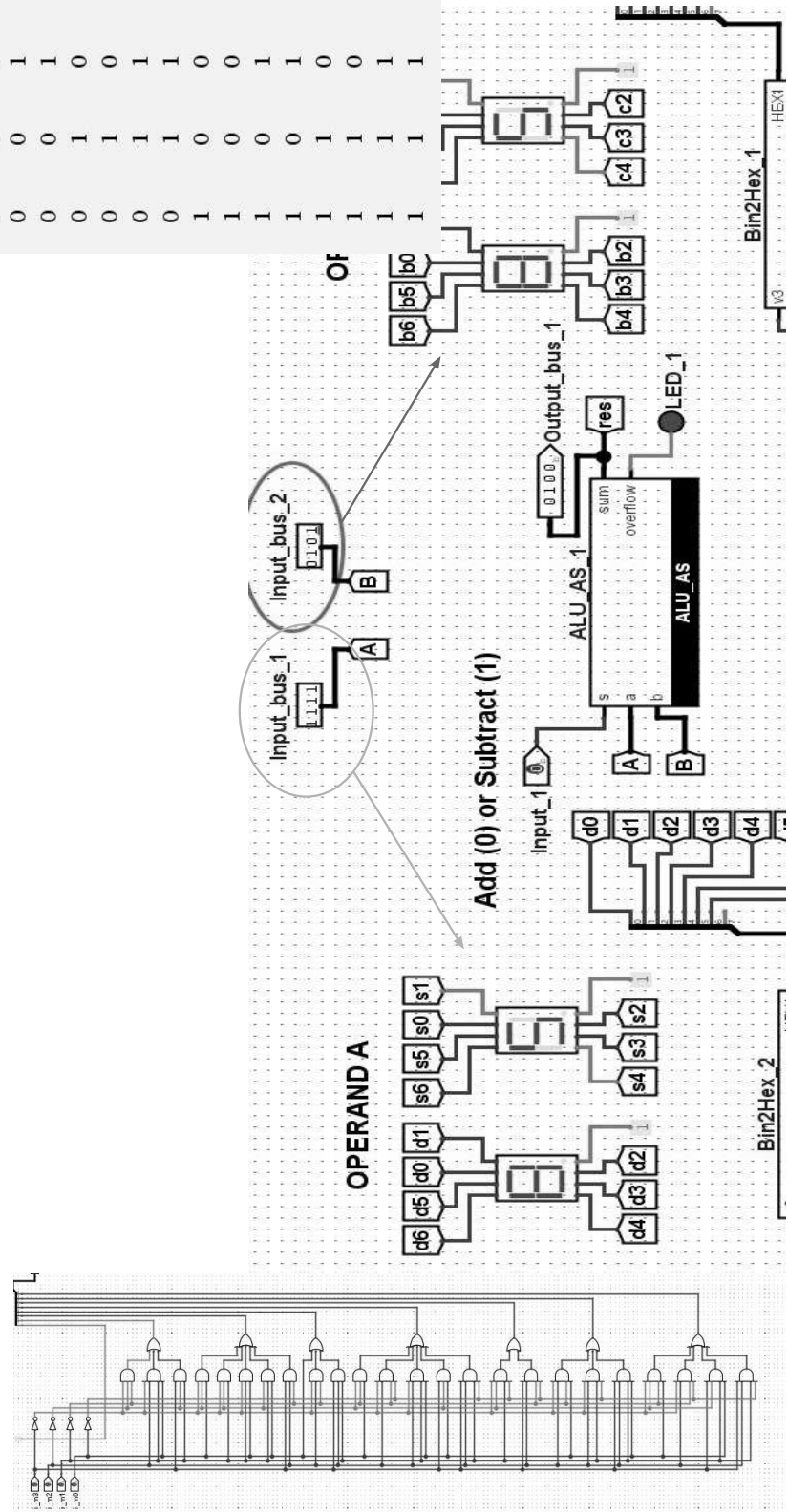


LAB 4

Digital Design

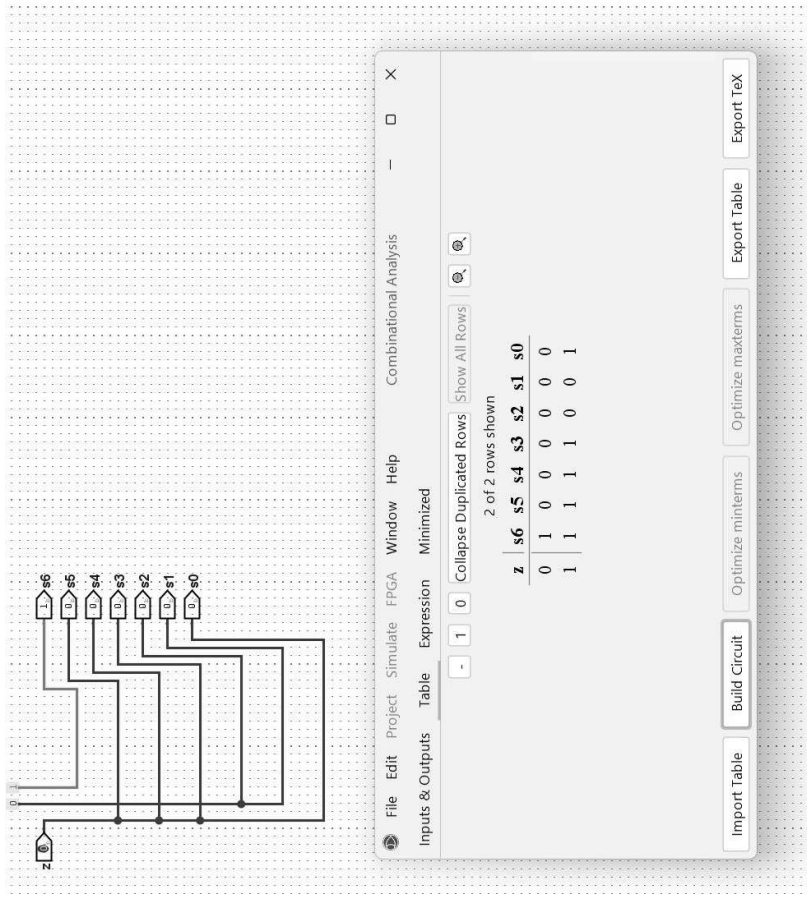
Eden Aklilu

First, we programmed the 7 Segment Display unit by putting in truth table values for numbers 1-15 (going back to 0 once we got to 10). Here is proof of it working (15 displayed on the left, 5 on the right)!! :)

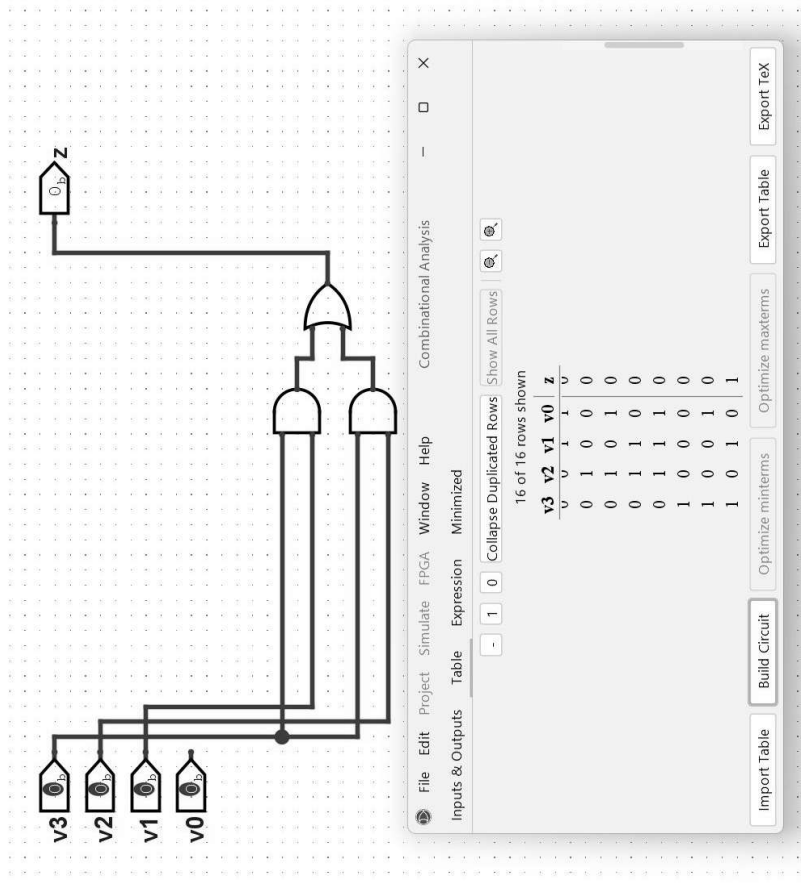


i	m3	i	m2	i	m1	i	m0	o	seg[7..0]
0	0	0	0	0	0	0	0	1	10000000
0	0	0	0	0	0	0	1	1	11110001
0	0	0	0	0	1	0	0	1	10100100
0	0	0	0	0	1	1	0	1	10110000
0	0	0	1	0	0	0	1	1	10010010
0	0	0	1	0	0	1	0	1	10010010
0	0	0	1	0	1	0	0	1	10000010
0	0	0	1	1	0	0	0	1	11111000
1	0	0	0	0	0	0	0	1	10000000
1	0	0	0	0	0	1	0	1	10011000
1	0	0	0	1	0	0	0	1	11000000
1	0	0	1	0	0	0	0	1	11111001
1	0	1	0	0	0	0	0	1	10100100
1	1	0	0	0	0	0	0	1	10110000
1	1	0	0	0	1	0	0	1	10011001
1	1	0	0	1	0	0	0	1	10010010

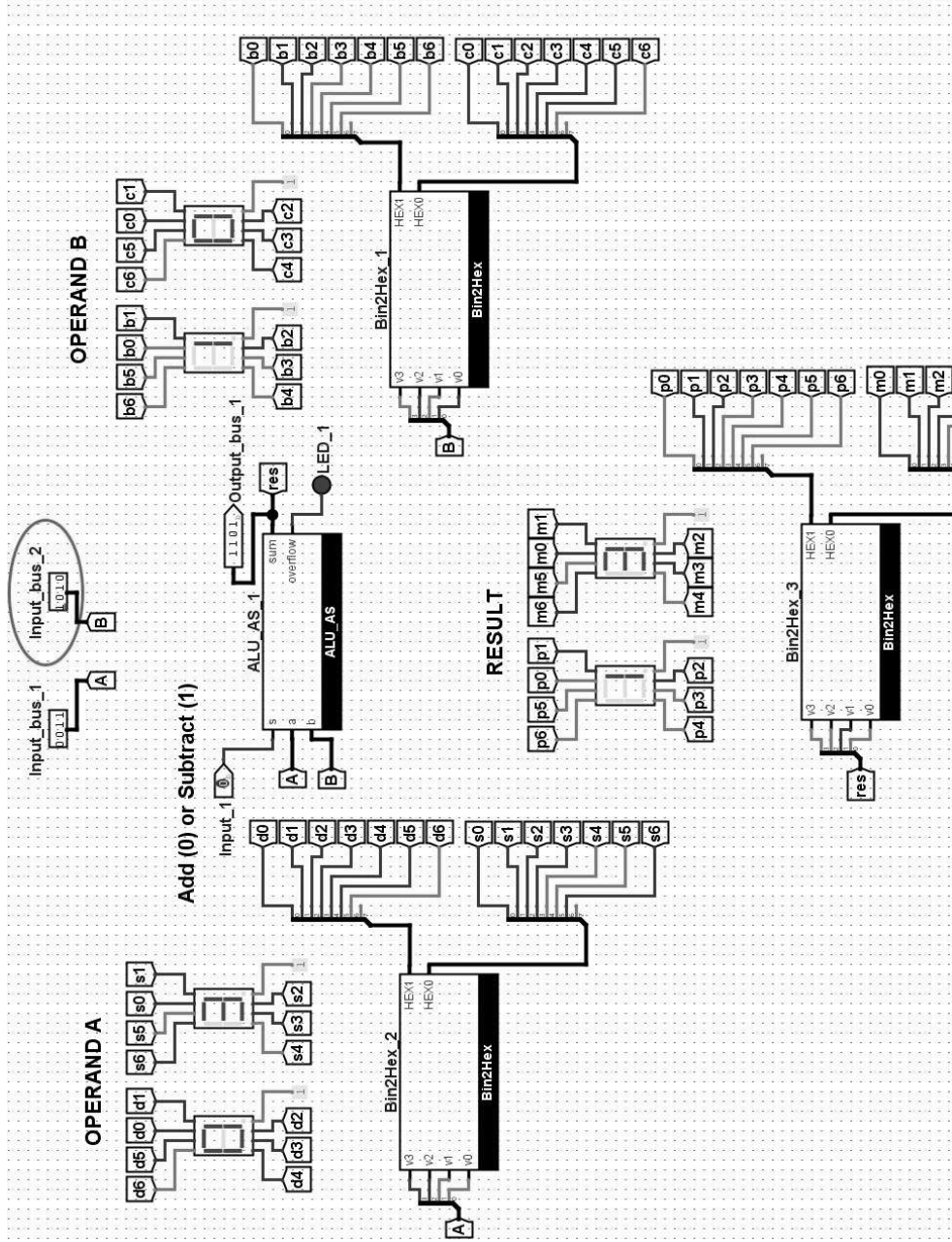
Circuit B for 0 or 1 (displays the first digit of the numbers)



Comparator (tells if the number is greater than 9 then pass a 1 through, which gives the first digit of a two digit number)



Results: It works!



Video proof as well, 9 + 5 and 9 - 5 worked!